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Primary Ideal

Notation 1.1. All rings will be unital and commutative.

Definition 1.2. Let R be a ring. An ideal $\mathfrak{q} \subseteq R$ is a **primary ideal** if for any $a, b \in R$ with $ab \in \mathfrak{q}$, either $a \in \mathfrak{q}$ or $b^n \in \mathfrak{q}$ for some $n \in \mathbb{Z}_{\geq 1}$. Equivalently, $\mathfrak{q} \subseteq R$ is a primary ideal of $R/\mathfrak{q}$ is a ring where every zero-divisor is nilpotent.

The radical of a primary ideal is necessarily a prime ideal. If $\mathfrak{q} \subseteq R$ is a primary ideal with radical $\mathfrak{p}$, then we say $\mathfrak{q}$ is a **p-primary ideal**.

Remark 1.3. Radical being prime does not implies the ideal is primary. For example, consider the ideal $I = (x^2, xy) \subseteq k[x, y]$ for some field k . Then $\sqrt{I} = (x)$ is prime, but I is not primary since $xy \in I$ but $x \notin I$ and $y^n \notin I$ for any $n \in \mathbb{Z}_{\geq 1}$. Another way to see this is that $xy = 0$ in R/I , so y is a zero divisor but it is not nilpotent.

However, radical being maximal is a sufficient condition for an ideal to be primary:

Lemma 1.4

Let R be a ring. If $\mathfrak{q} \subseteq R$ is an ideal such that $\mathfrak{m} := \sqrt{\mathfrak{q}}$ is a maximal ideal, then $\mathfrak{q}$ is a **m-primary ideal**.

Proof. It suffices to show that every zero-divisor in $R/\mathfrak{q}$ is nilpotent. Prime ideals of $R/\mathfrak{q}$ is in bijection with prime ideals of R containing $\mathfrak{q}$. Since radical of $\mathfrak{q}$ is the intersection of all prime ideals containing $\mathfrak{q}$, it follows that $\mathfrak{m} = \sqrt{\mathfrak{q}}$ implies the only prime ideal of R containing $\mathfrak{q}$ is $\mathfrak{m}$.

Thus, $\mathfrak{m}$ is the unique prime ideal of $R/\mathfrak{q}$. So $R/\mathfrak{q}$ is a local ring with maximal ideal $\mathfrak{m}$ and all zero-divisors in $R/\mathfrak{q}$ is contained in $\mathfrak{m}$ since any element outside $\mathfrak{m}$ is a unit. Now $\mathfrak{m} = \sqrt{\mathfrak{q}}$ implies the some power of any element of $\mathfrak{m}$ is in $\mathfrak{q}$, which implies some power of any zero-divisor in $R/\mathfrak{q}$ is zero, i.e. nilpotent. $\square$

Lemma 1.5

Let R be a ring and $S \subseteq R$ be a multiplicative subset. If R is a ring where every zero-divisor is nilpotent, then $S^{-1}R$ is also a ring where every zero-divisor is nilpotent.

Proof. Suppose $(a/s)(b/t) = 0/1$ in $S^{-1}R$, then $(ab)/(st) = 0/1$ implies there exists some $u \in S$ such that $u(ab) = 0$ in R . So a is a zero divisor in R and the assumption implies $a^n = 0$ for some $n \in \mathbb{Z}_{\geq 1}$. Then $a/s \in S^{-1}R$ is nilpotent because $(a/s)^n = a^n/s^n = 0/1$. $\square$

Theorem 1.6

Let R be a ring and $S \subseteq R$ be a multiplicative subset. If $\mathfrak{q} \subseteq R$ is a **p-primary ideal** and $\mathfrak{p} \cap S = \emptyset$, then $S^{-1}\mathfrak{q} \subseteq S^{-1}R$ is a **$S^{-1}\mathfrak{p}$ -primary ideal**.

Proof. ($S^{-1}\mathfrak{q}$ is a primary ideal) It suffices to show $S^{-1}R/S^{-1}\mathfrak{q}$ is a ring where every zero-divisor is nilpotent. Since localization is an exact functor, we have $S^{-1}R/S^{-1}\mathfrak{q} \cong S^{-1}(R/\mathfrak{q})$ as an $S^{-1}R$ -module. Both sides are also $S^{-1}R$ -algebras and the module isomorphism respect multiplication, so this is in fact

an isomorphism of rings. Then by lemma 1.5, we have

$$\begin{aligned} \mathfrak{q} \subseteq R \text{ is a primary ideal} &\implies R/\mathfrak{q} \text{ is a ring where every zero-divisor is nilpotent} \\ &\implies S^{-1}(R/\mathfrak{q}) \text{ is a ring where every zero-divisor is nilpotent} \\ &\implies S^{-1}R/S^{-1}\mathfrak{q} \text{ is a ring where every zero-divisor is nilpotent} \\ &\implies S^{-1}\mathfrak{q} \subseteq S^{-1}R \text{ is a primary ideal} \end{aligned}$$

$(\sqrt{S^{-1}\mathfrak{q}} = S^{-1}\mathfrak{p})$ $\sqrt{S^{-1}\mathfrak{q}}$ is the intersection of all prime ideal of $S^{-1}R$ containing $S^{-1}\mathfrak{q}$. So $\sqrt{S^{-1}\mathfrak{q}} \subseteq S^{-1}\mathfrak{p}$ because $S^{-1}\mathfrak{p}$ is a prime ideal of $S^{-1}R$ containing $S^{-1}\mathfrak{q}$. This uses the assumption that $S \cap \mathfrak{p} = \emptyset$.

Conversely, suppose $a/s \in S^{-1}\mathfrak{p}$ with $a \in \mathfrak{p}$ and $s \in S$. Then $a^n \in \mathfrak{q}$ for some $n \in \mathbb{Z}_{\geq 1}$ because $\mathfrak{p} = \sqrt{\mathfrak{q}}$. S is multiplicative and $s \in S$ implies $s^n \in S$. It follows that $(a/s)^n = a^n/s^n \in S^{-1}\mathfrak{q}$ which shows $a/s \in \sqrt{S^{-1}\mathfrak{q}}$. This completes the proof because every element of $S^{-1}\mathfrak{p}$ is a finite sum of a/s with $a \in \mathfrak{p}$ and $s \in S$. $\square$

Proposition 1.7

Let R be a ring and $S \subseteq R$ be a multiplicative subset. Let $\mathfrak{p}$ be a prime ideal of R with $\mathfrak{p} \cap S = \emptyset$. Then every $S^{-1}\mathfrak{p}$ -primary ideal is of the form $S^{-1}\mathfrak{q}$ where $\mathfrak{q} \subseteq R$ is a $\mathfrak{p}$ -primary ideal.

Proof. Let $\mathfrak{a}$ be a $S^{-1}\mathfrak{q}$ -primary ideal and define

$$\mathfrak{q} = \left\{ a \in R \mid \frac{a}{1} \in \mathfrak{a} \right\} \subseteq R$$

Then $\mathfrak{q} \subseteq R$ is an ideal and $\mathfrak{a} = S^{-1}\mathfrak{q}$.

To see $\mathfrak{q}$ is a $\mathfrak{p}$ -primary ideal, suppose $ab \in \mathfrak{q}$. Then $ab/1 = (a/1)(b/1) \in S^{-1}\mathfrak{q} = \mathfrak{a}$ which is a $S^{-1}\mathfrak{p}$ -primary ideal. So either $a/1 \in \mathfrak{a}$ or $(b/1)^n = b^n/1 \in \mathfrak{a}$ for some $n \in \mathbb{Z}_{\geq 1}$. Therefore, either $a \in \mathfrak{q}$ or $b^n \in \mathfrak{q}$ for some $n \in \mathbb{Z}_{\geq 1}$, which shows $\mathfrak{q}$ is a primary ideal. Finally, $\mathfrak{q}$ is a $\sqrt{\mathfrak{q}}$ -primary ideal implies $S^{-1}\mathfrak{q}$ is a $(S^{-1}\sqrt{\mathfrak{q}})$ -primary ideal by theorem 1.6. This shows

$$S^{-1}\sqrt{\mathfrak{q}} = \sqrt{S^{-1}\mathfrak{q}} = \sqrt{\mathfrak{a}} = S^{-1}\mathfrak{p}$$

Since prime ideal of $S^{-1}R$ is in bijection with prime ideal of R disjoint from S , we conclude $\sqrt{\mathfrak{q}} = \mathfrak{p}$. $\square$

Length of Module

Definition 2.1. Let R be a ring and M be an R -module. M is a **simple module** if $M \neq 0$ and the only submodule of M is zero and M itself. A **chain** of M is a sequence of submodules

$$M = M_0 \supsetneq M_1 \supsetneq M_2 \supsetneq \cdots \supsetneq M_n = 0$$

A **composition series** is a maximal chain of M , in particular, quotients between consecutive submodules are simple modules. The **length** of M is

$$\ell_R(M) = \inf\{n \in \mathbb{Z}_{\geq 0} \mid \text{there exist a composition series } M = M_0 \supsetneq M_1 \supsetneq \cdots \supsetneq M_n = 0\}$$

Lemma 2.2

Let R be a ring. If $N \subsetneq M$ is a strict inclusion of R -modules. Then either $\ell_R(N) = \ell_R(M) = \infty$ or $\ell_R(N) < \ell_R(M)$.

Proof. Let $M = M_0 \supsetneq M_1 \supsetneq M_2 \supsetneq \cdots \supsetneq M_n = 0$ be a composition series of M of minimal length. Let

$N_i = N \cap M_i$. Then

$$N = N_0 \supseteq N_1 \supseteq N_2 \supseteq \cdots \supseteq N_n = 0$$

Note that $N_i/N_{i+1} = (M_i \cap N)/(M_{i+1} \cap N) \subseteq M_i/M_{i+1}$. Thus N_i/N_{i+1} is either 0 or simple. If $N_i/N_{i+1} = M_i/M_{i+1}$ for all i , then $N = M$. Therefore, $N_i/N_{i+1} = 0$ for some i , which shows $\ell_R(N) < \ell_R(M)$. $\square$

Theorem 2.3

Let R be a ring and M be an R -module with a composition series of length n . Then

- (1) Every composition series of M has length n .
- (2) Every chain of M has length at most n .
- (3) Every chain of M can be extended to a composition series of M .

Proof. We may assume n is the smallest length of a composition series of M . Let $M = M_0 \supsetneq M_1 \supsetneq M_2 \supsetneq \cdots \supsetneq M_n = 0$ be the given composition series of M .

If $M = M'_0 \supsetneq M'_1 \supsetneq M'_2 \supsetneq \cdots \supsetneq M'_m = 0$ is another composition series of M , then $m \geq n$. On the other hand, $\ell(M'_1) \leq \ell(M) - 1$, $\ell(M'_2) \leq \ell(M) - 2$, $\ell(M'_3) \leq \ell(M) - 3$ and so on by lemma 2.2. Thus, $0 \leq \ell(M'_m) \leq \ell(M) - m = n - m$ which implies $m \leq n$. The remaining parts are clear. $\square$

Corollary 2.4: Length is Additive

Let R be a ring and let $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ be a short exact sequence of R -modules. Then

$$\ell_R(M_2) = \ell_R(M_1) + \ell_R(M_3).$$

Lemma 2.5

Let R be a ring and M be an R -module. The following are equivalent:

- (1) M is simple.
- (2) $\ell_R(M) = 1$.
- (3) $M \cong R/\mathfrak{m}$ for some maximal ideal $\mathfrak{m} \subseteq R$.

Proof. (3) $\implies$ (2) $\implies$ (1) is clear. For (1) $\implies$ (3), let $x \in M \setminus \{0\}$. The submodule generated by x is M itself because M is simple. Consider the R -module homomorphism $\varphi : R \rightarrow M$ defined by $r \mapsto rx$. φ is surjective because x generate M , and $\ker \varphi$ is a maximal ideal of R because M is simple and the image of any ideal containing $\ker \varphi$ would be a submodule of M . $\square$

Remark 2.6. In a local ring R with unique maximal ideal $\mathfrak{m}$, any simple R -module is isomorphic to the residue field $R/\mathfrak{m}$.

Lemma 2.7

Let $(R, \mathfrak{m}, k)$ be a local ring. If M is a finitely generated R -module annihilated by $\mathfrak{m}$ so that it can be viewed as an $k = R/\mathfrak{m}$ -module, then $\ell_R(M) = \dim_k M$.

Proof. Let $x_1, \dots, x_n$ be a basis of M as an k -module/vector space. The $x_1, \dots, x_n$ also generate M as an R -module because any element of M can be written as a R -linear combination of $x_1, \dots, x_n$ (write as a k -linear combination first, then lift the coefficients to R).

For any $i \in \{0, \dots, n\}$, let M_i denote the submodule generated by $x_{i+1}, \dots, x_n$ (for example $M_0 = M$, $M_n = 0$). Then $M = M_0 \supseteq M_1 \supseteq M_2 \supseteq \dots \supseteq M_n = 0$ is a chain of M and $M_i/M_{i+1} \cong k$ is generated by x_{i+1} . Since $\mathfrak{m}$ annihilates x_{n+1} , it follows that the R -module homomorphism

$$\begin{aligned} R &\longrightarrow M_i/M_{i+1} \\ r &\longmapsto rx_{i+1} \end{aligned}$$

is surjective with kernel containing $\mathfrak{m}$. So M_i/M_{i+1} is either zero or isomorphic to $R/\mathfrak{m} = k$. If $M_i/M_{i+1} = 0$, then x_{i+1} is in the R -submodule of M generated by $x_{i+2}, \dots, x_n$, contradicting the k -linear independence of $x_1, \dots, x_n$.

Thus M_i/M_{i+1} is isomorphic to k which implies they are simple R -modules by lemma 2.5. So $M = M_0 \supseteq M_1 \supseteq M_2 \supseteq \dots \supseteq M_n = 0$ is a composition series of M and $\ell_R(M) = n = \dim_k M$ by theorem 2.3. $\square$

Lemma 2.8

A noetherian local ring $(R, \mathfrak{m})$ has finite length (as an R -module) if and only if $\mathfrak{m}^n = 0$ for some $n \in \mathbb{Z}_{\geq 1}$.

Proof. If $\mathfrak{m}^n \neq 0$ for all $n \in \mathbb{Z}_{\geq 1}$, then $\ell_R(R) = \infty$ because $R \supsetneq \mathfrak{m} \supsetneq \mathfrak{m}^2 \supsetneq \mathfrak{m}^3 \supsetneq \dots$ is an infinite descending chain of submodules. Each inclusion is strict because if $\mathfrak{m}^n = \mathfrak{m}^{n+1}$, then $\mathfrak{m}^n = 0$ by Nakayama's lemma.

Conversely, if $\mathfrak{m}^n = 0$ for some $n \in \mathbb{Z}_{\geq 1}$, then consider the chain $R \supsetneq \mathfrak{m} \supsetneq \mathfrak{m}^2 \supsetneq \dots \supsetneq \mathfrak{m}^{n-1} \supsetneq \mathfrak{m}^n = 0$. Each quotient $\mathfrak{m}^i/\mathfrak{m}^{i+1}$ is finitely generated because R is noetherian and is annihilated by $\mathfrak{m}$. By lemma 2.7, $\mathfrak{m}^i/\mathfrak{m}^{i+1}$ have finite length. So we can extend the chain to a composition series of R with finitely many terms. Therefore, R has finite length. $\square$

Definition 2.9. Let $(R, \mathfrak{m})$ be a local ring. A **parameter ideal for R** is an ideal $\mathfrak{q} \subseteq \mathfrak{m}$ such that $R/\mathfrak{q}$ has finite length.

Lemma 2.10

Let $(R, \mathfrak{m})$ be a noetherian local ring. An ideal $\mathfrak{q} \subseteq \mathfrak{m}$ is a parameter ideal for R if and only if $\mathfrak{m}^n \subseteq \mathfrak{q}$ for some $n \in \mathbb{Z}_{\geq 1}$.

Proof. $R/\mathfrak{q}$ is a local ring with maximal ideal $\mathfrak{m}/\mathfrak{q}$. So $R/\mathfrak{q}$ has finite length if and only if $(\mathfrak{m}/\mathfrak{q})^n = 0$ for some $n \in \mathbb{Z}_{\geq 1}$ by lemma 2.8. Now the ideal $(\mathfrak{m}/\mathfrak{q})^n$ in the ring $R/\mathfrak{q}$ is precisely $\mathfrak{m}^n/\mathfrak{q}$. Hence the condition that $(\mathfrak{m}/\mathfrak{q})^n = 0$ is equivalent to $\mathfrak{m}^n/\mathfrak{q} = 0$ which is equivalent to $\mathfrak{m}^n \subseteq \mathfrak{q}$. $\square$

Corollary 2.11

Let $(R, \mathfrak{m})$ be a noetherian local ring. Then $\mathfrak{q}$ is a parameter ideal for R if and only if $\mathfrak{q}$ is a $\mathfrak{m}$ -primary ideal.

Proof. R is noetherian implies $\mathfrak{m}$ is finitely generated. By lemma 2.10 and lemma 1.4, we have

$$\begin{aligned} \mathfrak{q} \text{ is a parameter ideal for } R &\iff \mathfrak{m}^n \subseteq \mathfrak{q} \text{ for some } n \in \mathbb{Z}_{\geq 1} \\ &\iff \sqrt{\mathfrak{q}} = \mathfrak{m} \\ &\iff \mathfrak{q} \text{ is a } \mathfrak{m}\text{-primary ideal} \end{aligned}$$

□

Corollary 2.12

Let k be a field and $R = k[x_1, \dots, x_n]$ be a polynomial ring over k . Let $\mathfrak{m}$ be the maximal ideal generated by the variables $x_1, \dots, x_n$. Then the parameter ideals for the local ring $R_{\mathfrak{m}}$ are precisely the localizations of $\mathfrak{m}$ -primary ideals of R .

Proof. Parameter ideals for $R_{\mathfrak{m}}$ are $\mathfrak{m}R_{\mathfrak{m}}$ -primary ideals of $R_{\mathfrak{m}}$ by corollary 2.11. Every $\mathfrak{m}R_{\mathfrak{m}}$ -primary ideal of $R_{\mathfrak{m}}$ is the localization of a $\mathfrak{m}$ -primary ideal of R by proposition 1.7. □

Hilbert-Samuel Polynomial

Goals:

- (1) Prove some result saying length can be computed by dimension over some field in some situations.
- (2) Prove some result that expresses the Hilbert-Samuel polynomial of a module over some local ring in terms of things Macaulay 2 can compute.

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